Supplementary Material: Task-Relevant Ultrasonic Information Acquisition for Impacted Composites

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1 S1. Evidence authority and statistical units

The supplement exposes detailed controls, sensitivities, implementation diagnostics, and machine-readable provenance omitted from the compressed main paper. Numerical authority remains the 39-row canonical metrics file. Physical specimens are the first statistical unit and the six held-out experimental domains are equally weighted. State, checkpoint, candidate, seed, and learner- pair rows are repeated computational records rather than independent samples. All teacher, oracle, and component-substitution quantities are retrospective unless an evidence record explicitly marks them deployable.

The codebase refers to the supervised state-conditioned closed-loop implementation as MAVIS. This is one implementation of State-Conditioned Task-Oriented Acquisition, not the paper-level method. The code identifier `mvd_m1_o2` denotes the static deployable reference used in the final comparison; it is not a published competing method.

Record group	Content
S01–S04	Full domain and checkpoint records for spatial, sparse, static, and conditional-value comparisons.
S05–S09	Real-state, acquired-position/history, reconstruction, shuffled-content, and dynamic-valuation controls.
S10–S12	Component substitutions and bounded reachable-set planning records.
S13–S16	Learner-pair value stability, feedback strata, and implementation diagnostics.
S17–S18	Source hashes, chronology, and external-data feasibility boundaries.

2 S2. Spatial information and sparse recoverability

The registered main comparison uses the matched B-family scalar-to-spatial path. The distinct independent internal-field estimator (`U1_INDEPENDENT_FIELD_SENSITIVITY`) had point MAE 0.08964 and is retained only as a sensitivity path. It is not substituted for the matched 32.1% effect.

The sparse condition observes a registered 25% normalized-raster pattern and uses bilinear interpolation before the frozen field pathway. Its 89.9% retention is referenced to the matched full-field gain. The residual sparse-to- full MAE gap remains positive in all domains, and the sampling fraction has no scanner-time interpretation.

3 S3. Task-conditioned and predictor-conditioned value

The cross-objective oracle result is bounded by the learned global-mask test. Source-trained global mechanics and reconstruction masks produced the predeclared support indicator 0; the learned global mechanics mask therefore did not reproduce the oracle separation. The reconstruction task in this comparison is the registered normalized-RGB-MSE objective, not a claim about all image representations or metrics.

Under identical strict-OOF states and splits, Ridge–Huber best-action agreement was 0.675 (95% CI 0.622–0.727) (`U5_RIDGE_HUBER_BEST_ACTION`), and top-k agreement was 0.685 (95% CI 0.650–0.719) (`U5_RIDGE_HUBER_TOPK`). Ridge–MLP best-action agreement was 0.233 (95% CI 0.182–0.283) (`U5_RIDGE_MLP_BEST_ACTION`), and top-k agreement was 0.213 (95% CI 0.192–0.235) (`U5_RIDGE_MLP_TOPK`). Because the shallow MLP was substantially less accurate, these contrasts do not isolate predictor architecture at equal predictive accuracy.

4 S4. State-conditioned valuation and source controls

The registered exact-budget set regrets were 0.08171 for the static scorer (`O1_STATIC_SET_REGRET`), 0.07993 for the global score (`O1_GLOBAL_SET_REGRET`), and 0.07978 for the random median (`O1_RANDOM_SET_REGRET`). These source-only references annotate the static representation; they do not establish information-theoretic unobservability.

At the endpoint, the real-state trajectory recovered 26.6% (95% CI 10.6%–41.0%) of the static-to-independent-full-field error ratio (`O3_FULL_FIELD_RECOVERY`). Its denominator is the independent internal-field sensitivity estimator, not the matched B-family field path.

The acquired-position/history control inherits the exact positions, ordering, and costs of the recipient trajectory while removing measured content. The reconstruction control uses the registered image objective on the same legal trajectory. The shuffled-content control preserves recipient positions and budget while sourcing values from separated donor content. These semantics are required when interpreting the adverse real-minus-control directions.

5 S5. Cost-constrained realization and implementation diagnostics

The reachable-set analysis is restricted to the registered two-action pool at 6.25%. The joint near-oracle compares reachable sets retrospectively; it is not deployable and does not establish arbitrary-horizon planning regret.

For `A3_FEEDBACK_BENEFIT`, the no-feedback reference retained an AUEBC advantage of 1.496×10^{-5} (95% CI 1.064×10^{-5}–1.944×10^{-5}). With the registered sign convention, feedback benefit was -1.496×10^{-5} (95% CI -1.944×10^{-5} to -1.064×10^{-5}), with feedback favorable in two of six held-out domains. This is a frozen implementation diagnostic and is not generalized to other state representations or policies.

For `A4_BASELINE_MINUS_MAVIS`, equal-domain CAI AUEBC was 0.125053 for the supervised state-conditioned implementation and 0.124992 for the static reference. Reference minus learned was -6.114×10^{-5} (95% CI -8.461×10^{-5} to -3.777×10^{-5}), and the learned implementation improved two of six held-out domains. The direction favors the reference. This comparison is one system-level endpoint and does not define the performance of the full Task- Relevant Information Acquisition framework.

6 S6. Provenance, chronology, and scope boundaries

The evidence chronology separates pre-existing spatial and static-reference analyses, the frozen outer implementation endpoint, and later diagnostic analyses. Later diagnostics reused fixed states, source bindings, splits, and outcomes; they were not used to select or modify the frozen endpoint. No model training, tuning, result selection, or canonical numerical claim was changed for the manuscript identity reframe.

The machine-readable visibility map assigns every canonical claim to main headline, main support, one main system diagnostic, or supplement-only use. Source-hash records bind every number to its immutable result artifact, and the supplement manifest independently records copied-file digests. External-data records are feasibility audits only and contain no external method-performance evaluation.

Scope remains the paired 276-specimen, six-domain CFRP data program; exact native-raster digital cost; the fixed predictor families, state representations, action hierarchy, and planners; and the registered CAI and normalized-RGB-MSE tasks. Physical path length, coupling, settling, hardware timing, industrial cost, and external empirical transfer are not established.